AGA KHAN UNIVERSITY EXAMINATION BOARD

SECONDARY SCHOOL CERTIFICATE

CLASS X EXAMINATION

MAY 2013

Physics Paper II

Time: 2 hours 25 minutes Marks: 40

INSTRUCTIONS

Please read the following instructions carefully.

1. Check your name and school information. Sign if it is accurate.

I agree that this is my name and school. **Candidate's signature**

2. RUBRIC. There are NINE questions. Answer ALL questions. Questions 7, 8 & 9 each offer TWO choices. Attempt any ONE choice from each.

3. When answering the questions:

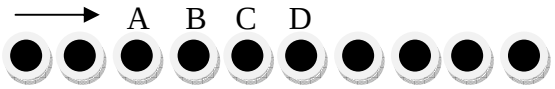
   Read each question carefully.
   Use a black pencil for diagrams. DO NOT use coloured pencils.
   DO NOT use staples, paper clips, glue, correcting fluid or ink erasers.
   Complete your answer in the allocated space only. DO NOT write outside the answer box.

4. The marks for the questions are shown in brackets ( ).

5. You may use a simple calculator if you wish.

Q.1. (Total 4 Marks)

A, B, C and D in the given figure represent particles in a medium through which waves are passing continuously in the direction indicated by the arrows.



Describe how these particles will move through the following types of waves.

i. Transverse waves

_____

_____

_____

_____

ii. Longitudinal waves

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Q.2. (Total 5 Marks)

a. State the laws of refraction. (2 Marks)

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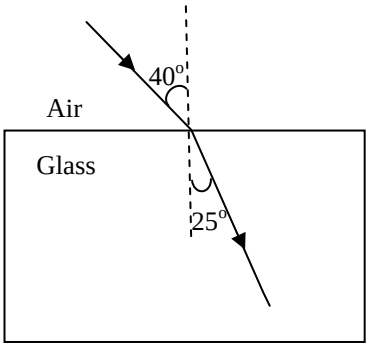
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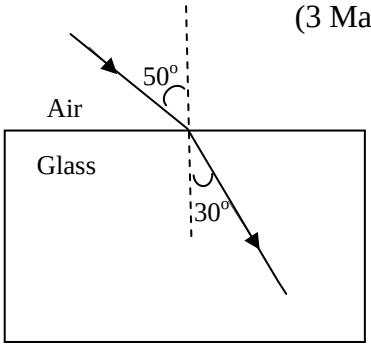
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b. Find the values of refractive index for the given diagrams and also calculate their average. (3 Marks)

Angle (θ)	Sine θ
25°	0.4226
30°	0.5000
40°	0.6428
50°	0.7660





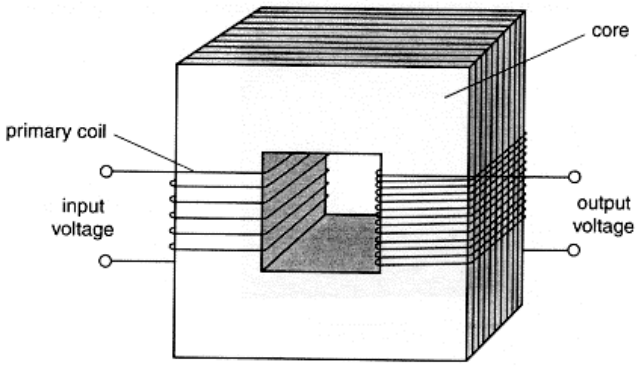
Q.3. (Total 6 Marks)

a. State and explain Coulomb's law. (3 Marks)

b. Find the force between the two electrostatic charges of 200×10^{-6} coulomb and 300×10^{-6} coulomb respectively placed in air, 0.003 m apart from each other. (3 Marks)

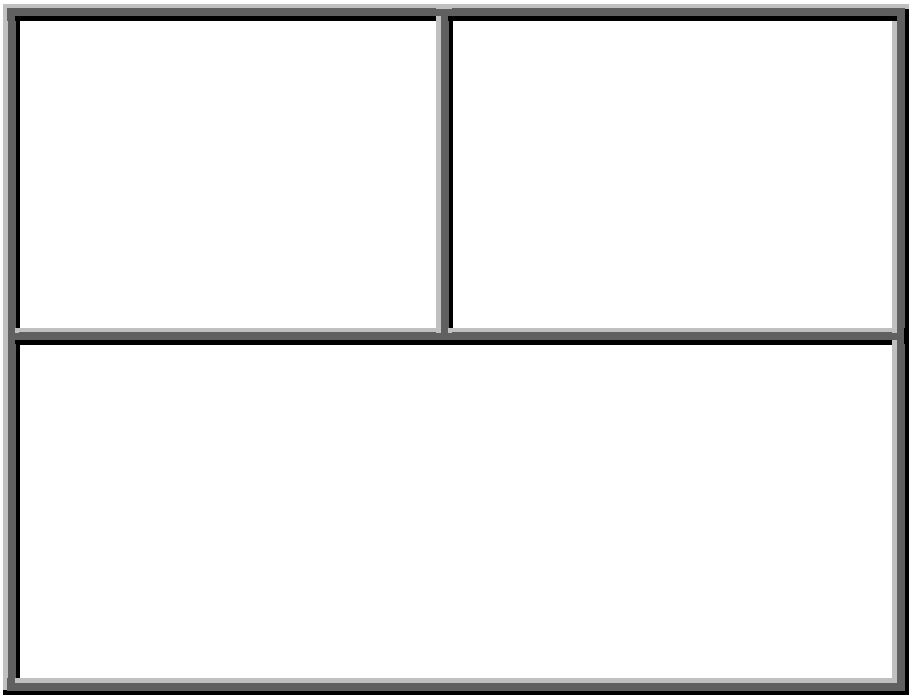
Q.4. (Total 3 Marks)

The given diagram shows a simple transformer. What type of transformer is it and what changes can be made so that voltage may be reduced to a low output?



Q.5. (Total 3 Marks)

Draw the symbol of NOT, OR and AND gates.



Q.6. (Total 4 Marks)

The given table provides some information about a radioactive element.

Half life	Total Time for Decay	Initial Mass of the Given Sample	Remaining Mass
1600 years	8000 years	X	25 gm

Calculate the value of 'X'.

Q.7. (Total 5 Marks)

EITHER

a. How are mechanical and electromagnetic waves different from each other? Describe in FIVE points.

OR

b. Unwanted noise can be a source of stress and irritation for many people. It also affects reading ability and concentration of school going children.

Discuss that noise pollution has become a threat for our global living environment. Support your answer by mentioning some major sources of noise pollution.



Q.8. (Total 5 Marks)

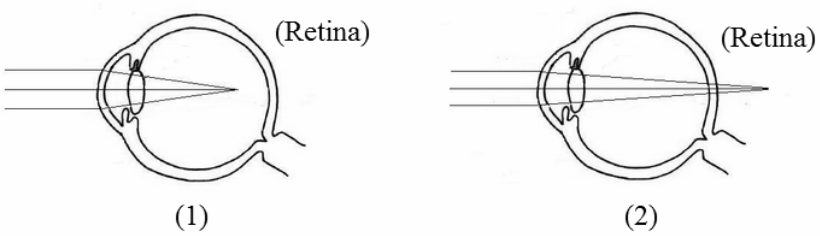
EITHER

a. Complete the given table, when an object is placed before a concave mirror at different position.

Position of object	Position of image	Nature of image	Size of image
	Between ‘F’ and ‘C’		
At ‘C’			
	Beyond ‘C’		
At ‘F’			
Between ‘F’ and ‘C’			

OR

b. The given diagrams show defects of a human eye.



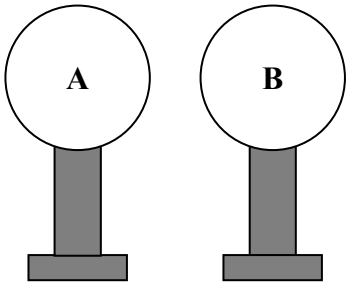
(1) (2)

Identify the defects in the above diagrams. Why are the rays not converging on the retina in both cases? Explain how these defects can be corrected.

Q.9. (Total 5 Marks)

EITHER

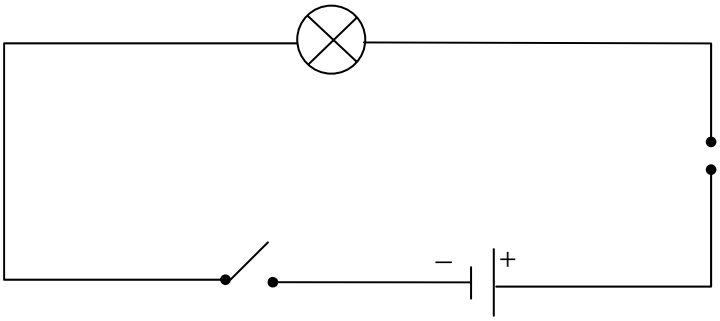
a. In the given diagram, two metal spheres ‘A’ and ‘B’ are fixed on wooden stands and they are in contact with each other.



Explain how the metal spheres ‘A’ and ‘B’ can be charged, using a negatively charged ebonite rod without touching them. Also provide diagrammatic illustration of the procedure.

OR

b. Name the type of circuit given below and describe all the components used in the circuit.



END OF PAPER

Please use this page for rough work